TOR AMPLITUDE IS UNCHANGED BY PASSING TO THE SMALL ETALE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete, covering both directions of (1) and deriving (2) formally

Editorial note. The label lemma-tor-dimension-rel names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the localization functor epsilon^* from D(O_X) to the small etale derived category over f^{-1}O_Y
- tor amplitude of complexes over a morphism of ringed spaces, checked on stalks
- stalks of pullback sheaves at geometric points and strict henselization
- faithful flatness of henselization and its invariance for tor amplitude

Source reference. Original lemma 09U9: sources/stacks-chow/tags.txt, source lines 12417–12417. Prerequisite pointer invoked at proof line(s) 3281; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 08H7: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 320–320. Prerequisite pointer invoked at proof line(s) 3237; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04HX: sources/stacks-chow/tags.txt, source lines 5883–5883. Prerequisite pointer invoked at proof line(s) 3256; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03Q1: sources/stacks-chow/tags.txt, source lines 4878–4878. Prerequisite pointer invoked at proof line(s) 3255; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DK0: sources/stacks-chow/tags.txt, source lines 16995–16995. Prerequisite pointer invoked at proof line(s) 3268; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07QM: sources/stacks-chow/tags.txt, source lines 9837–9837. Prerequisite pointer invoked at proof line(s) 3274; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DJF: sources/stacks-chow/tags.txt, source lines 16975–16975. Prerequisite pointer invoked at proof line(s) 3278; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 008H: sources/stacks-chow/tags.txt, source lines 317–317. Prerequisite pointer invoked at proof line(s) 3247; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08H5: sources/stacks-chow/tags.txt, source lines 10768–10768. Prerequisite pointer invoked at proof line(s) 3242; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let E be an object of $D(\mathcal{O}_X)$. Then*

1. *E as an object of $D(f^{-1}\mathcal{O}_Y)$ has tor amplitude in $[a, b]$ if and only if ϵ^*E has tor amplitude in $[a, b]$ as an object of $D(f^{-1}_{small}\mathcal{O}_{Y_{étale}})$.*
2. *E locally has finite tor dimension as an object of $D(f^{-1}\mathcal{O}_Y)$ if and only if ϵ^*E locally has finite tor dimension as an object of $D(f^{-1}_{small}\mathcal{O}_{Y_{étale}})$.*

Here ϵ is as in (08H7).

Proof. The easy direction in (1) follows from Cohomology on Sites, Lemma 08H5. Let $x \in X$ be a point and let $\overline{x}$ be a geometric point lying over x. Let $y = f(x)$ and denote $\overline{y}$ the geometric point of Y coming from $\overline{x}$. Then $(f^{-1}\mathcal{O}_Y)_x = \mathcal{O}_{Y,y}$ (Sheaves, Lemma 008H) and

$$(f^{-1}_{small}\mathcal{O}_{Y_{étale}})_{\overline{x}} = \mathcal{O}_{Y_{étale},\overline{y}} = \mathcal{O}^{sh}_{Y,y}$$

is the strict henselization (by Étale Cohomology, Lemmas 03Q1 and 04HX). Since the stalk of $\mathcal{O}_{X_{étale}}$ at X is $\mathcal{O}^{sh}_{X,x}$ we obtain

$$(\epsilon^*E)_{\overline{x}} = E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} \mathcal{O}^{sh}_{X,x}$$

by transitivity of pullbacks. If ϵ^*E has tor amplitude in $[a, b]$ as a complex of $f^{-1}_{small}\mathcal{O}_{Y_{étale}}$-modules, then $(\epsilon^*E)_{\overline{x}}$ has tor amplitude in $[a, b]$ as a complex of $\mathcal{O}^{sh}_{Y,y}$-modules (because taking stalks is a pullback and lemma cited above). By More on Flatness, Lemma 0DK0 we find the tor amplitude of $(\epsilon^*E)_{\overline{x}}$ as a complex of $\mathcal{O}_{Y,y}$-modules is in $[a, b]$. Since $\mathcal{O}_{X,x} \to \mathcal{O}^{sh}_{X,x}$ is faithfully flat (More on Algebra, Lemma 07QM) and since $(\epsilon^*E)_{\overline{x}} = E_x \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} \mathcal{O}^{sh}_{X,x}$ we may apply More on Algebra, Lemma 0DJF to conclude the tor amplitude of E_x as a complex of $\mathcal{O}_{Y,y}$-modules is in $[a, b]$. By Cohomology, Lemma 09U9 we conclude that E as an object of $D(f^{-1}\mathcal{O}_Y)$ has tor amplitude in $[a, b]$. This gives the reverse implication in (1). Part (2) follows formally from (1). □